

A Unified Gray Hammer–Rosen Model for Surface- and Bath-Temperature Drives

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July 15, 2026

Abstract

A compact Hammer–Rosen-type model is developed for planar gray, two-temperature Marshak waves. The model allows independent Rosseland and absorption opacity power laws and can be driven either by a prescribed surface radiation temperature $T_s(t)$ or by a prescribed bath temperature $T_{\text{bath}}(t)$ through a Marshak boundary condition. The radiation profile uses the first-order Hammer–Rosen curvature correction. The material profile is the same radiation shape multiplied by a monotone factor $\exp[-s(t)y]$, where the single scalar $s(t)$ is obtained from an areal material-energy closure. The surface-driven model requires two scalar physical ODEs, three running time quadratures, and one scalar root. The bath-driven model adds only one scalar ODE for $T_s(t)$. No spatial mesh, diffusion matrix, PDE solve, or history-dependent integral equation is required. A fully self-contained implementation recipe is given in the final section.

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1 Gray diffusion problem

Consider a planar, fixed-density material occupying $x > 0$. Hydrodynamic motion is neglected. The radiation energy density and material energy density are

$$E_r = a_{\text{rad}} T_r^4, \quad U = f \rho^{1-\mu} T^\beta, \quad a_{\text{rad}} = \frac{4\sigma_{\text{SB}}}{c}. \quad (1.1)$$

The Rosseland and absorption opacities are independent power laws,

$$\frac{1}{\kappa_R} = g T^\alpha \rho^{-\lambda}, \quad \frac{1}{\kappa_a} = g' T^{\alpha'} \rho^{-\lambda'}. \quad (1.2)$$

Thus (α, λ, g) are the Rosseland parameters and (α', λ', g') are the absorption parameters.

The underlying gray diffusion equations are

$$\frac{\partial}{\partial t} (a_{\text{rad}} T_r^4) = \frac{\partial}{\partial x} \left[\frac{cg}{3} \rho^{-1-\lambda} T^\alpha \frac{\partial}{\partial x} (a_{\text{rad}} T_r^4) \right] - \frac{a_{\text{rad}} c}{g'} \rho^{1+\lambda'} T^{-\alpha'} (T_r^4 - T^4), \quad (1.3)$$

$$\frac{\partial}{\partial t} (f \rho^{1-\mu} T^\beta) = \frac{a_{\text{rad}} c}{g'} \rho^{1+\lambda'} T^{-\alpha'} (T_r^4 - T^4). \quad (1.4)$$

The reduced model derived below does not solve (1.3)–(1.4) pointwise.

Two boundary specifications are considered.

Surface-temperature drive.

$$T_r(0, t) = T_s(t), \quad (1.5)$$

where $T_s(t)$ is prescribed.

Bath-temperature drive. The surface radiation temperature is unknown and the incoming bath is imposed through the Marshak condition

$$F_s(t) = \frac{a_{\text{rad}} c}{2} [T_{\text{bath}}^4(t) - T_s^4(t)], \quad T_s(t) = T_r(0, t), \quad (1.6)$$

where $F_s > 0$ denotes flux into the material.

Define the two standard Hammer–Rosen parameters

$$\varepsilon = \frac{\beta}{4 + \alpha}, \quad C = \frac{4a_{\text{rad}} c g}{3(4 + \alpha) f} \rho^{-2+\mu-\lambda}. \quad (1.7)$$

The exponent of the leading front profile will always be written explicitly as $1/(4 + \alpha - \beta)$; no separate symbol is introduced for it.

2 Boundary-drive structure

The formulas below use the instantaneous surface radiation temperature $T_s(t) = T_r(0, t)$. Its status must therefore be fixed before the common gray model is described.

2.1 Prescribed surface-temperature drive

For a Dirichlet radiation-temperature drive, $T_s(t)$ is known input data. The reduced model advances the surface material temperature, the areal material energy, and three running time integrals. The derivative of the prescribed $T_s(t)$ is used only in the first-order HR curvature coefficient.

2.2 Prescribed bath-temperature drive

For a Marshak bath drive, $T_{\text{bath}}(t)$ is the input and $T_s(t)$ is unknown. It is important that the bath problem is *not* organized as a separate conversion

$$T_{\text{bath}}(t) \longrightarrow T_s(t)$$

followed by an independent gray calculation. The boundary temperature depends on the current surface material temperature and on the accumulated HR front history, both of which themselves depend on $T_s(t)$. Therefore $T_s(t)$ is added to the ODE state and is integrated simultaneously with the common gray states.

At every ODE-stage evaluation, the time integrator supplies a current trial value of $T_s(t)$. That current value is used in the surface material ODE, the gray storage factor, and the base-front history. After those instantaneous quantities are evaluated, the Marshak/HR boundary equation supplies $\dot{T}_s(t)$ and the curvature coefficient $B(t)$. Thus all formulas that contain $T_s(t)$ remain explicit at an ODE stage even though $T_s(t)$ is not prescribed.

After $T_s(t)$ and $B(t)$ have been specified by the appropriate boundary branch, the interior material-lag closure and the final-front correction are identical for the two drive types.

3 Surface material temperature

Let

$$T_{m,0}(t) = T(0, t) \quad (3.1)$$

be the surface material temperature. In the surface-driven branch, $T_s(t)$ is read from the prescribed drive. In the bath-driven branch, the current trial value of $T_s(t)$ is supplied by the coupled ODE state described in Section 2. Evaluating the local material balance at $x = 0$ gives

$$\frac{d}{dt} [f \rho^{1-\mu} T_{m,0}^\beta(t)] = \frac{a_{\text{rad}}^C}{g'} \rho^{1+\lambda'} T_{m,0}^{-\alpha'}(t) [T_s^4(t) - T_{m,0}^4(t)]. \quad (3.2)$$

Equivalently,

$$\dot{T}_{m,0}(t) = \frac{a_{\text{rad}}^C}{\beta f g'} \rho^{\mu+\lambda'} T_{m,0}^{1-\alpha'-\beta}(t) [T_s^4(t) - T_{m,0}^4(t)]. \quad (3.3)$$

This equation naturally gives $T_{m,0} < T_s$ during heating and may give $T_{m,0} > T_s$ during cooling.

Equation (3.2) is one scalar, generally stiff ODE. In Python it should be integrated with `solve_ivp`, using `method="Radau"` or `method="BDF"`. At a perfectly cold start, a code may internally evolve the regular power $T_{m,0}^{\alpha'+\beta}$, but that auxiliary variable is only a numerical device and is not part of the analytical model.

4 Common-shape gray Hammer–Rosen front

4.1 Gray storage coordinate

For the leading common-shape reduction, write locally

$$T_r(x, t) = T_s(t)q(x, t), \quad T(x, t) = T_{m,0}(t)q(x, t), \quad 0 \leq q \leq 1. \quad (4.1)$$

The ratio of radiation energy to material energy at the surface is

$$R(t) = \frac{a_{\text{rad}} T_s^4(t)}{f \rho^{1-\mu} T_{m,0}^\beta(t)}. \quad (4.2)$$

At an arbitrary profile value q , the corresponding ratio is $R(t)q^{4-\beta}$. The gray storage coordinate entering the HR reduction is therefore

$$Z(q; R) = (4 + \alpha - \beta) \int_0^q \frac{\xi^{3+\alpha-\beta}}{1 + R\xi^{4-\beta}} d\xi. \quad (4.3)$$

The surface value $Z(1; R)$ is the gray storage factor

$$G(R) = (4 + \alpha - \beta) \int_0^1 \frac{\xi^{3+\alpha-\beta}}{1 + R\xi^{4-\beta}} d\xi. \quad (4.4)$$

The denominator is the normalized total storage $1 + R\xi^{4-\beta}$; thus G reduces the effective drive whenever radiation energy is non-negligible.

The integral has the exact Gauss-hypergeometric form

$$G(R) = {}_2F_1\left(1, \frac{4+\alpha-\beta}{4-\beta}; 1 + \frac{4+\alpha-\beta}{4-\beta}; -R\right). \quad (4.5)$$

The identity follows from

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a} du, \quad (4.6)$$

valid initially for $\Re c > \Re b > 0$ and elsewhere by analytic continuation. In Python, (4.5) is evaluated directly with `hyp2f1`.

4.2 Base front

Define temporarily the running history

$$\mathcal{I}(t) = \int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'. \quad (4.7)$$

Applying the first-order HR front moment to the gray storage coordinate gives

$$x_{F,0}^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C T_{m,0}^{-\beta}(t) \mathcal{I}(t). \quad (4.8)$$

Equivalently, without introducing $\mathcal{I}$,

$$x_{F,0}^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C T_{m,0}^{-\beta}(t) \int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'. \quad (4.9)$$

Numerically this is one scalar running quadrature. It is advanced together with the ODE states; no past time array needs to be reintegrated at every step.

5 First-order radiation profile and the two boundary drives

The normalized first-order HR radiation profile is

$$T_r(x, t) = T_s(t) \left[\left(1 - \frac{x}{x_F(t)}\right) \left(1 + B(t) \frac{x}{x_F(t)}\right) \right]^{1/(4+\alpha-\beta)}. \quad (5.1)$$

The coefficient $B(t)$ is the first HR curvature correction. The profile requires $B(t) > -1$ so that its base remains nonnegative on the wave interval.

5.1 Prescribed surface radiation temperature

For prescribed $T_s(t)$, the HR correction is

$$B(t) = \frac{\varepsilon}{2} \left[1 - \frac{2 + \varepsilon}{1 - \varepsilon} \frac{\int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'}{T_s^4(t) T_{m,0}^\alpha(t) G(R(t))} \times \frac{d}{dt} \ln T_s^{4+\alpha}(t) \right]. \quad (5.2)$$

This is an algebraic post-processing formula once the base-front history is known. For a tabulated drive, use a shape-preserving interpolant such as `scipy.interpolate.PchipInterpolator`; its analytical derivative can be obtained with `PchipInterpolator.derivative()`. If the tabulated pulse contains measurement noise, regularize only this derivative, not the drive used in the energy quadratures.

5.2 Prescribed bath temperature

At the surface, the diffusion flux predicted by (5.1), using the base front inside the boundary closure, is

$$F_s(t) = \frac{4a_{\text{rad}}cg}{3(4 + \alpha - \beta)x_{F,0}(t)} \rho^{-1-\lambda} T_{m,0}^\alpha(t) T_s^4(t) [1 - B(t)]. \quad (5.3)$$

Equating this to the Marshak flux (1.6) gives

$$B(t) = 1 - \frac{3(4 + \alpha - \beta)x_{F,0}(t)\rho^{1+\lambda}}{8gT_{m,0}^\alpha(t)} \left[\frac{T_{\text{bath}}^4(t)}{T_s^4(t)} - 1 \right]. \quad (5.4)$$

Using $x_{F,0}$ rather than the final moment-corrected front keeps the boundary subsystem sequential; replacing it by x_F would introduce a higher-order implicit feedback.

Equating (5.4) to the HR curvature relation gives one scalar ODE for the unknown surface radiation temperature:

$$\dot{T}_s(t) = \frac{2T_s^5(t)T_{m,0}^\alpha(t)G(R(t))}{\varepsilon(4 + \alpha) \frac{2 + \varepsilon}{1 - \varepsilon} \int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'} \times \left\{ \frac{3(4 + \alpha - \beta)x_{F,0}(t)\rho^{1+\lambda}}{8gT_{m,0}^\alpha(t)} \left[\frac{T_{\text{bath}}^4(t)}{T_s^4(t)} - 1 \right] - 1 + \frac{\varepsilon}{2} \right\}. \quad (5.5)$$

The bath drive therefore adds only one physical scalar ODE. It uses $T_{\text{bath}}(t)$ but not $\dot{T}_{\text{bath}}(t)$, which is advantageous for tabulated data. Equation (5.5) is not solved before the gray model: it is integrated simultaneously with the surface material ODE, the base-front history, and the areal material-energy ODE. At each stage of `solve_ivp`, the solver provides the current values of T_s , $T_{m,0}$, and the running history integral; these determine R , $G(R)$, $x_{F,0}$, B , and finally $\dot{T}_s$.

The cold onset is formally singular because the base-front history vanishes; a robust implementation starts at the first positive drive time with a small positive thermal seed and checks insensitivity to that seed.

After the current $T_s(t)$ and $B(t)$ have been obtained from the selected boundary branch, the interior equations are identical.

6 Monotone material-profile closure

The material profile is represented by the same radiation shape multiplied by one monotone ratio factor,

$$T(x, t) = T_{m,0}(t) \left[\left(1 - \frac{x}{x_F(t)} \right) \left(1 + B(t) \frac{x}{x_F(t)} \right) \right]^{1/(4+\alpha-\beta)} \times \exp \left[-s(t) \frac{x}{x_F(t)} \right]. \quad (6.1)$$

The ratio is

$$\frac{T(x, t)}{T_r(x, t)} = \frac{T_{m,0}(t)}{T_s(t)} \exp \left[-s(t) \frac{x}{x_F(t)} \right]. \quad (6.2)$$

Positive s narrows the material profile during heating, $s = 0$ gives proportional profiles, and negative s permits retained material energy during cooling. This parametrization remains positive for every real s and does not change the leading front exponent.

7 Profile moments and their closed forms

Every spatial integral required by the reduced model can be generated from two moments,

$$M_p(t, z) = \int_0^1 [(1-y)(1+B(t)y)]^{p/(4+\alpha-\beta)} e^{-zy} dy, \quad (7.1)$$

$$N_p(t, z) = \int_0^1 y [(1-y)(1+B(t)y)]^{p/(4+\alpha-\beta)} e^{-zy} dy. \quad (7.2)$$

These integrals have exact forms in terms of Humbert's confluent hypergeometric function Φ_1 :

$$M_p(t, z) = \frac{1}{1 + \frac{p}{4 + \alpha - \beta}} \Phi_1 \left(1, -\frac{p}{4 + \alpha - \beta}; 2 + \frac{p}{4 + \alpha - \beta}; -B(t), -z \right), \quad (7.3)$$

$$N_p(t, z) = \frac{1}{\left(1 + \frac{p}{4 + \alpha - \beta} \right) \left(2 + \frac{p}{4 + \alpha - \beta} \right)} \Phi_1 \left(2, -\frac{p}{4 + \alpha - \beta}; 3 + \frac{p}{4 + \alpha - \beta}; -B(t), -z \right). \quad (7.4)$$

The original integral representation of Humbert's function is

$$\Phi_1(A, B; C; x, z) = \frac{\Gamma(C)}{\Gamma(A)\Gamma(C-A)} \int_0^1 e^{zu} u^{A-1} (1-u)^{C-A-1} (1-xu)^{-B} du. \quad (7.5)$$

The conditions relevant here are $B(t) > -1$ and $p/(4 + \alpha - \beta) > -1$.

SciPy provides `hyp2f1` for $G(R)$ but does not currently expose Humbert Φ_1 directly. Although Φ_1 can be represented by nested series of one-variable hypergeometric functions, that is not the recommended production route. The most transparent and accurate Python evaluation of (7.1)–(7.2) is `quad`, which uses adaptive QUADPACK integration and returns an error estimate. The algebraic endpoint behavior can be supplied through `weight="alg"`. If performance later matters, a fixed Gauss–Jacobi rule can be validated against `quad`.

8 Areal material-energy equation and algebraic recovery of $s(t)$

The material energy per unit transverse area represented by the reduced profile is

$$U_m(t) = x_{F,0}(t) f \rho^{1-\mu} T_{m,0}^\beta(t) M_\beta(t, \beta s(t)). \quad (8.1)$$

The base front $x_{F,0}$ is used in this closure so that the material-energy subsystem remains sequential.

Integrating the material equation (1.4) over the reduced profile gives

$$\dot{U}_m(t) = \frac{a_{\text{rad}} c}{g'} \rho^{1+\lambda'} x_{F,0}(t) \left[T_s^4(t) T_{m,0}^{-\alpha'}(t) M_{4-\alpha'}(t, -\alpha' s(t)) - T_{m,0}^{4-\alpha'}(t) M_{4-\alpha'}(t, (4-\alpha') s(t)) \right]. \quad (8.2)$$

Here $U_m(t)$ is an ordinary ODE state. It is not known in advance, just as $T_{m,0}(t)$ is not known in advance. At each internal stage of the ODE solver, the solver supplies a current trial value of $U_m(t)$. With that current value, determine $s(t)$ from

$$M_\beta(t, \beta s(t)) = \frac{U_m(t)}{x_{F,0}(t) f \rho^{1-\mu} T_{m,0}^\beta(t)}. \quad (8.3)$$

The recovered $s(t)$ is then inserted immediately into the right-hand side of (8.2). Consequently, the ODE solver sees a closed scalar equation for U_m : the scalar root is merely an internal algebraic operation required to evaluate $\dot{U}_m$.

There is no circular time dependence. The numerical order at a single ODE stage is

1. read the current trial value of U_m from the ODE state;
2. solve (8.3) for the current s ;
3. evaluate the two exchange moments in (8.2);
4. return $\dot{U}_m$ to the ODE integrator.

The left side of (8.3) is strictly decreasing because

$$\frac{\partial}{\partial s} M_\beta(t, \beta s) = -\beta \int_0^1 y [(1-y)(1+B(t)y)]^{\beta/(4+\alpha-\beta)} e^{-\beta s y} dy < 0. \quad (8.4)$$

Hence the root is unique. In Python, use the bracketed solver `brentq`. This is a local scalar inversion, not a Volterra equation; all material-energy history is compressed into the single ODE state $U_m(t)$.

A natural initialization at the first positive-drive time is the common-shape state $s = 0$,

$$U_m(t_{\text{start}}) = x_{F,0}(t_{\text{start}}) f \rho^{1-\mu} T_{m,0}^\beta(t_{\text{start}}) M_\beta(t_{\text{start}}, 0). \quad (8.5)$$

At an exactly cold start, both $x_{F,0}$ and $T_{m,0}$ vanish and the normalized root equation is singular; the first-positive-drive initialization avoids this coordinate singularity.

It is possible to differentiate (8.1) and derive an ODE directly for $s(t)$, but that introduces $\dot{x}_{F,0}$, $\dot{T}_{m,0}$, and $\dot{B}$. For a prescribed surface drive, $\dot{B}$ can require a second derivative of the drive. Evolving U_m and recovering s algebraically is therefore simpler and substantially more robust.

9 Final front correction

For any trial material-lag value s , define the first total-energy moment

$$\boxed{\mathcal{E}_1(t, s) = a_{\text{rad}} T_s^4(t) N_4(t, 0) + f \rho^{1-\mu} T_{m,0}^\beta(t) N_\beta(t, \beta s).} \quad (9.1)$$

Define also the dimensionless transport integrand

$$\boxed{\mathcal{D}(t, s) = T_s^4(t) T_{m,0}^\alpha(t) \left\{ [1 - B(t)] M_\beta(t, \alpha s) + 2B(t) N_\beta(t, \alpha s) \right\}.} \quad (9.2)$$

The compact form follows from the identity

$$\begin{aligned} & [(1-y)(1+By)]^{(\alpha+3)/(4+\alpha-\beta)} \left[-\frac{d}{dy} ((1-y)(1+By))^{1/(4+\alpha-\beta)} \right] \\ &= \frac{(1-B) + 2By}{4+\alpha-\beta} [(1-y)(1+By)]^{\beta/(4+\alpha-\beta)}. \end{aligned} \quad (9.3)$$

All dimensional constants in the physical transport moment are identical in the numerator and denominator below and therefore cancel.

The final moment-corrected front is

$$\boxed{x_F^2(t) = x_{F,0}^2(t) \frac{\mathcal{E}_1(t, 0)}{\mathcal{E}_1(t, s(t))} \frac{\int_0^t \mathcal{D}(t', s(t')) dt'}{\int_0^t \mathcal{D}(t', 0) dt'}.} \quad (9.4)$$

The denominator is simply the special case $s = 0$ of the same integrand. Numerically the two time integrals are two running scalar quadratures. Once $x_F(t)$ is known, the temperature profiles follow algebraically from (5.1) and (6.1).

10 Relation to LTE Hammer–Rosen theory

10.1 The LTE manifold of the gray model

Strong absorption coupling corresponds to a large exchange rate $(c/g')\rho^{1+\lambda'}T^{-\alpha'}$. The reduced model is then driven toward the LTE manifold

$$\boxed{T_{m,0}(t) = T_s(t), \quad s(t) = 0.} \quad (10.1)$$

The first equality makes the surface material and radiation temperatures equal. The second equality follows from the unique root (8.3): on the common-profile LTE manifold, the areal material energy is proportional to $M_\beta(t, 0)$, so uniqueness forces $s = 0$.

On this manifold,

$$\mathcal{E}_1(t, s) = \mathcal{E}_1(t, 0), \quad \mathcal{D}(t, s) = \mathcal{D}(t, 0), \quad (10.2)$$

and the final moment correction becomes unity:

$$\boxed{x_F(t) = x_{F,0}(t).} \quad (10.3)$$

The gray base front reduces to

$$\boxed{x_F^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C T_s^{-\beta}(t) \int_0^t T_s^{4+\alpha}(t') G(R(t')) dt',} \quad (10.4)$$

with

$$R(t) = \frac{a_{\text{rad}}}{f\rho^{1-\mu}} T_s^{4-\beta}(t). \quad (10.5)$$

Equation (10.4) is an LTE HR-type model retaining radiation-energy storage through $G(R)$.

The classical material-dominated HR theory is recovered when $R \ll 1$. Since

$$G(R) = 1 + \mathcal{O}(R), \quad R \rightarrow 0, \quad (10.6)$$

one obtains the standard result

$$\boxed{x_F^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C T_s^{-\beta}(t) \int_0^t T_s^{4+\alpha}(t') dt'.} \quad (10.7)$$

The surface-driven expression for $B(t)$ likewise reduces to the usual first-order LTE HR curvature correction. Therefore the prescribed- T_s gray model recovers the well-known LTE HR solution in the combined limit of strong material-radiation coupling and material-dominated storage.

10.2 $T_{\text{bath}} = T_s$ is not the LTE limit

LTE means $T = T_r$ inside the material. It does not mean that the bath and surface radiation temperatures are equal. The Marshak condition is

$$F_s(t) = \frac{a_{\text{rad}} c}{2} [T_{\text{bath}}^4(t) - T_s^4(t)]. \quad (10.8)$$

Consequently,

$$T_{\text{bath}}(t) = T_s(t) \implies F_s(t) = 0. \quad (10.9)$$

Thus setting $T_{\text{bath}} = T_s$ in the bath-driven branch gives zero net incident flux; it does not reproduce a prescribed- T_s Dirichlet boundary. A driven Marshak wave normally has $T_{\text{bath}} > T_s$. The frequently used substitution “ $T_s = T_{\text{bath}}$ in the surface-HR formula” is a separate naive approximation, not a limit of the Robin/Marshak boundary condition.

10.3 Comparison with the Cohen–Heizler bath-temperature construction

The standard LTE HR theory assumes that $T_s(t)$ is known. For a bath-driven LTE problem, Cohen and Heizler Appendix A closes the unknown surface temperature by combining the Marshak flux with the global HR areal-energy relation. In the material-dominated LTE notation,

$$n = 4 + \alpha, \quad x_F^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C T_s^{-\beta}(t) \int_0^t T_s^n(t') dt', \quad (10.10)$$

and the HR areal energy is

$$E_{\text{HR}}(t) = f\rho^{1-\mu}(1-\varepsilon)x_F(t)T_s^\beta(t). \quad (10.11)$$

The bath boundary then requires

$$\boxed{\frac{dE_{\text{HR}}}{dt} = \frac{a_{\text{rad}} c}{2} [T_{\text{bath}}^4(t) - T_s^4(t)]}. \quad (10.12)$$

Equations (10.10)–(10.12) form a coupled surface-temperature/history problem. Appendix A advances this boundary-energy system in time. The compact LTE fluence model developed previously replaces that time-marching closure by two bath fluences and one algebraic root, at the cost of assuming that the surface-to-bath ratio varies slowly over the accumulated history.

The present direct gray bath closure is different. It obtains the surface flux from the local gradient of the first-order HR radiation profile and equates that flux directly to the Marshak condition. This gives the scalar ODE (5.5) for $T_s(t)$ while retaining the non-LTE variables $T_{m,0}(t)$ and $s(t)$. In its LTE limit, it is a compact differential bath-HR closure with the same physical Marshak boundary and the same leading HR profile class as the Cohen–Heizler construction, but it is not generally algebraically identical to Appendix A. The two closures use different reduced information:

- Cohen–Heizler uses the global HR areal-energy relation and its time derivative;
- the present model uses the local HR surface gradient, the base front, and the first-order curvature relation.

Exact equivalence should therefore not be claimed without a separate asymptotic derivation. Their agreement is a numerical and modeling question. The direct closure is attractive because it extends without structural change to the gray two-temperature problem and requires no derivative of the prescribed bath history.

Table 1: Relation among the surface- and bath-driven LTE/gray reductions.

Model	Boundary input	Boundary calculation
Classical LTE HR	$T_s(t)$	No boundary solve; one HR history quadrature.
Cohen–Heizler	$T_{\text{bath}}(t)$	Coupled Marshak/global-HR energy time march for the unknown $T_s(t)$.
Appendix A		
Compact LTE fluence model	$T_{\text{bath}}(t)$	Two bath fluences and one scalar root; uses a fluence-ratio approximation.
Unified gray HR, surface branch	$T_s(t)$	Adds surface material and areal material-energy ODEs plus the local s root.
Unified gray HR, bath branch	$T_{\text{bath}}(t)$	Adds one simultaneous scalar ODE for $T_s(t)$ using the direct Marshak/HR surface-gradient closure.

11 Numerical implementation and complexity

11.1 Recommended SciPy functions

Table 2: Recommended Python functions for the reduced model.

Task	Function	Practical recommendation
Scalar ODE system	<code>solve_ivp</code>	Use <code>Radau</code> or <code>BDF</code> ; request dense output if profiles are needed at specified times.
Gray storage $G(R)$	<code>hyp2f1</code>	Evaluate the exact closed form (4.5); compare once against <code>quad</code> as a unit test.
Profile moments M_p, N_p	<code>quad</code>	Use tight <code>epsabs/epsrel</code> ; optionally use <code>weight="alg"</code> for the endpoint factor.
Scalar inversion for s	<code>brentq</code>	Bracket the unique root and use the previous accepted s value to construct a small initial bracket.
Tabulated drive interpolation	<code>PchipInterpolator</code>	For prescribed T_s , use its analytical interpolant derivative in $B(t)$. For prescribed T_{bath} , only the interpolated value is required.

11.2 Surface-temperature drive

For prescribed $T_s(t)$, the ODE solver advances $T_{m,0}(t)$ and $U_m(t)$ together with three running time-integral states: the base-front history and the two transport histories in (9.4). At every ODE-stage evaluation:

1. read $T_s(t)$ from the prescribed interpolant and evaluate its derivative only for $B(t)$;
2. read the current ODE states $T_{m,0}(t)$ and $U_m(t)$ and the three current running-integral values;
3. evaluate $R(t)$ and $G(R(t))$, then calculate $x_{F,0}(t)$ from the current base-front history;
4. calculate $B(t)$ from (5.2);
5. solve (8.3) for the current $s(t)$ using the current $U_m(t)$;
6. evaluate $T_{m,0}$, $\dot{U}_m$, and the derivatives of the three running integrals;
7. return this fixed-dimensional right-hand side to `solve_ivp`.

The final front and profiles are evaluated algebraically only at requested output times. The surface-driven model requires

$$\boxed{2 \text{ scalar physical ODEs} + 3 \text{ running time quadratures} + 1 \text{ local scalar root.}} \quad (11.1)$$

Each right-hand-side evaluation also calls a small number of one-dimensional profile integrals M_p, N_p ; these are adaptive function evaluations, not a spatial discretization or time-history solve.

11.3 Bath-temperature drive

For prescribed $T_{\text{bath}}(t)$, the ODE state additionally contains the unknown $T_s(t)$. It is not calculated first. At each ODE-stage evaluation:

1. read the prescribed $T_{\text{bath}}(t)$ and the current trial states $T_s(t), T_{m,0}(t), U_m(t)$;
2. evaluate $R(t)$, $G(R(t))$, and $x_{F,0}(t)$ from the current states and base-front history;
3. evaluate $B(t)$ from the Marshak relation (5.4) and $\dot{T}_s(t)$ from (5.5);
4. solve the same unique root (8.3) using the current $U_m(t)$;
5. evaluate $\dot{T}_{m,0}$, $\dot{U}_m$, and the three running-integral derivatives;
6. return the coupled scalar right-hand side to the same stiff ODE solver.

No derivative of the bath history is needed. The bath-driven model requires

$$\boxed{3 \text{ scalar physical ODEs} + 3 \text{ running time quadratures} + 1 \text{ local scalar root.}} \quad (11.2)$$

The state dimension is fixed and independent of spatial resolution.

By comparison, a full gray PDE calculation requires hundreds or thousands of spatial cells, two unknown fields per cell, temperature-dependent face coefficients, an implicit matrix solve at every timestep, nonlinear opacity updates, boundary-face treatment, timestep control, and spatial convergence studies. The reduced model performs only a small fixed number of scalar operations and one-dimensional profile integrations per ODE evaluation.

12 Numerical illustration

The model was tested for the supplied SiO_2 drive using both surface and bath boundary specifications. The simulations use a validated $N = 1000$ finite-volume gray diffusion solver without a flux limiter. The bath-driven simulation uses the Marshak Robin boundary condition directly.

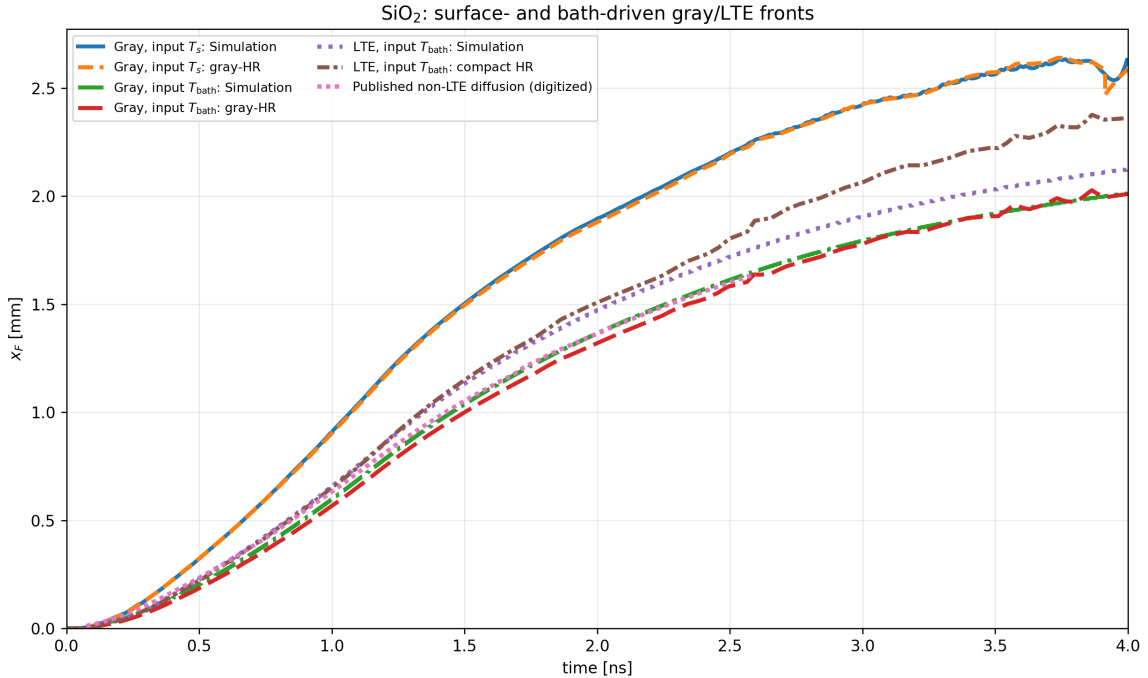


Figure 1: Surface- and bath-driven gray/LTE front comparisons through 4 ns. The unified gray-HR model uses the same interior closure for both gray drive types; only the boundary subsystem differs.

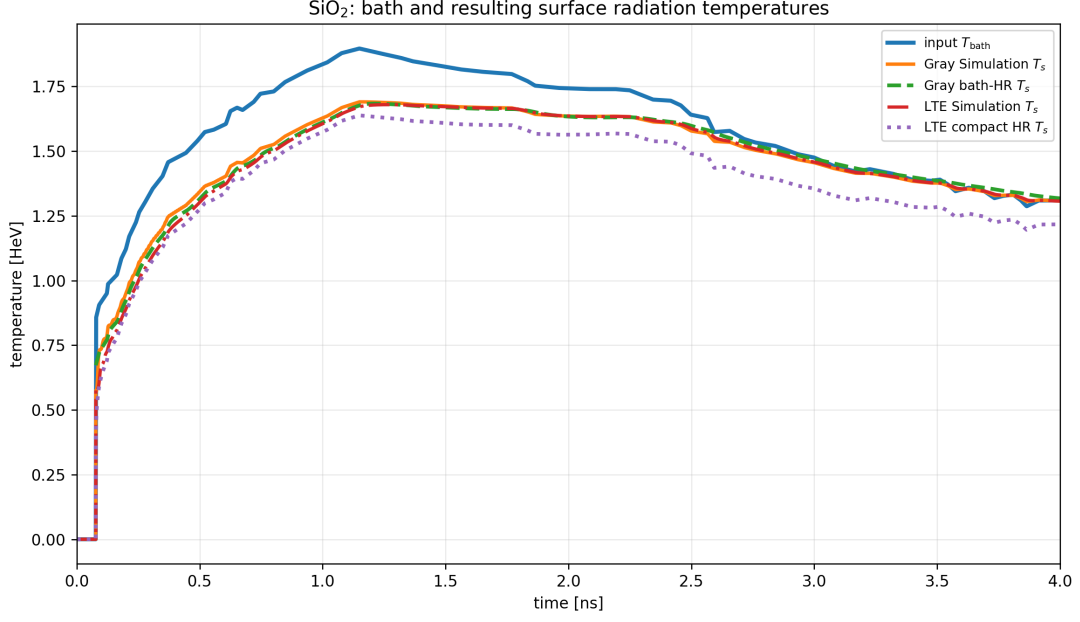


Figure 2: Prescribed bath temperature and the resulting surface radiation temperatures. The direct gray bath-HR boundary ODE follows the gray Robin simulation closely.

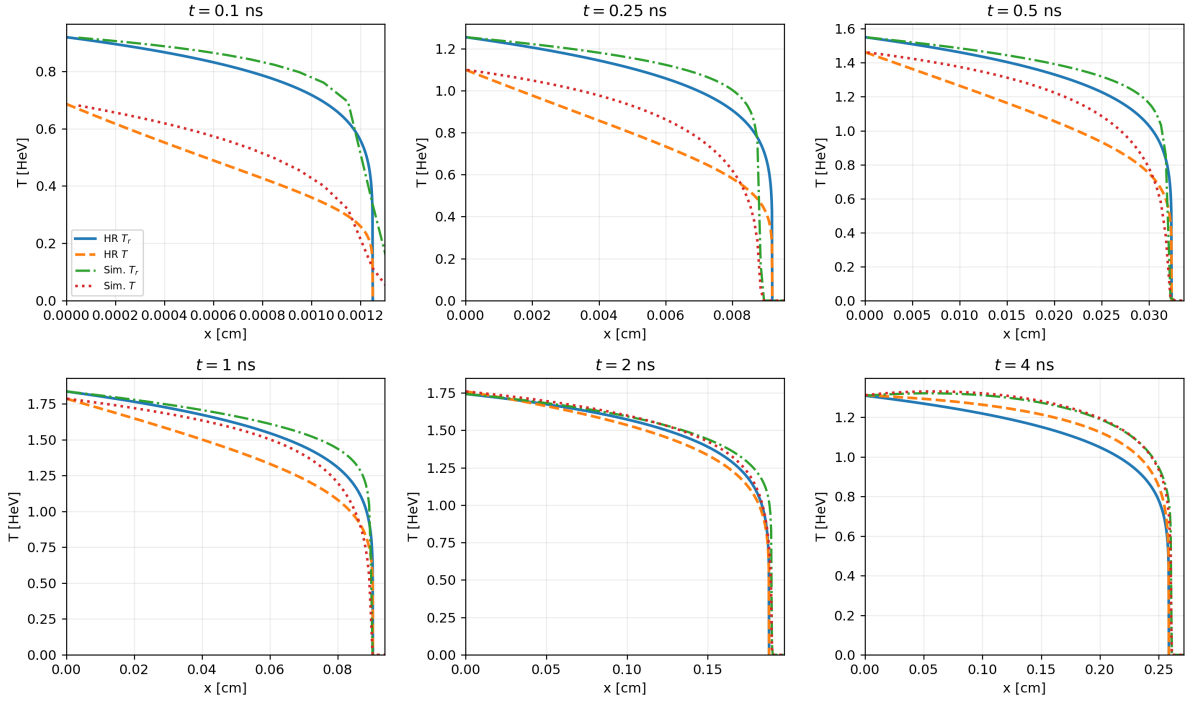


Figure 3: Gray radiation and material profiles for the surface-driven example. The monotone factor $\exp[-s(t)y]$ captures the early material narrowing without changing the front exponent.

For the bath-driven gray case, the front error decreases from approximately 8% at 0.5 ns to less than 1% near 4 ns. For the surface-driven gray case, the front remains near the one-percent level over most of the pulse. The larger relative errors immediately after the discontinuous turn-on occur inside the short onset layer where both the HR asymptotics and a numerical front

definition are least reliable.

13 Accuracy and limitations

The model combines a first-order HR radiation profile with integrated material energy and transport moments. It is not a pointwise solution of the gray PDE and is not a proof of a uniform $\mathcal{O}(\varepsilon)$ expansion for arbitrary drives. Its principal approximations are:

- one HR-shaped radiation profile;
- one scalar material-lag parameter $s(t)$;
- use of the base front $x_{F,0}$ inside the surface-flux and material-energy closures;
- enforcement of integrated moments rather than the PDEs pointwise.

The model is expected to work best for sharp, single-front waves with smooth power-law material properties. Strong opacity transitions, multiple fronts, large excursions with $B(t) \leq -1$, or strongly non-HR radiation profiles indicate that another profile parameter or the full gray PDE may be required.

14 Self-contained summary of the working model

This section is an independent implementation specification. No auxiliary profile symbol is introduced; the temperature profiles and all profile moments are written explicitly.

Problem definition, material laws, and drive choice

A fixed-density material occupies $x > 0$. The gray radiation and material energy densities are

$$E_r(x, t) = a_{\text{rad}} T_r^4(x, t), \quad U(x, t) = f \rho^{1-\mu} T^\beta(x, t), \quad \boxed{a_{\text{rad}} = \frac{4\sigma_{\text{SB}}}{c}}. \quad (14.1)$$

The Rosseland and absorption opacity laws are

$$\frac{1}{\kappa_R} = g T^\alpha \rho^{-\lambda}, \quad \frac{1}{\kappa_a} = g' T^{\alpha'} \rho^{-\lambda'}. \quad (14.2)$$

Define

$$\boxed{\varepsilon = \frac{\beta}{4 + \alpha}, \quad C = \frac{4a_{\text{rad}} c g}{3(4 + \alpha) f} \rho^{-2+\mu-\lambda}}, \quad (14.3)$$

$$\boxed{T_{m,0}(t) = T(0, t), \quad T_s(t) = T_r(0, t)}. \quad (14.4)$$

Choose one boundary branch before evaluating the common formulas:

A. Prescribed surface radiation temperature. $T_s(t)$ is input data. Interpolate it with `PchipInterpolator`; its derivative is required only in the curvature formula below.

B. Prescribed bath temperature. $T_{\text{bath}}(t)$ is input data and $T_s(t)$ is an additional unknown ODE state. It is integrated simultaneously with $T_{m,0}(t)$, $U_m(t)$, and the running time integrals. At every ODE stage, the solver supplies the current trial value of $T_s(t)$, so all common formulas below are evaluated using that current value. The derivative $\dot{T}_s(t)$ is supplied by the bath equation in Step 3. There is no separate $T_{\text{bath}} \rightarrow T_s$ preprocessing calculation.

The Marshak boundary convention is

$$\boxed{F_s(t) = \frac{a_{\text{rad}} c}{2} [T_{\text{bath}}^4(t) - T_s^4(t)]}. \quad (14.5)$$

Step 1: surface material temperature — one scalar ODE

Using the prescribed $T_s(t)$ in branch A, or the current ODE-stage value of $T_s(t)$ in branch B, solve

$$\boxed{\frac{d}{dt} [f \rho^{1-\mu} T_{m,0}^\beta(t)] = \frac{a_{\text{rad}} c}{g'} \rho^{1+\lambda'} T_{m,0}^{-\alpha'}(t) [T_s^4(t) - T_{m,0}^4(t)]}. \quad (14.6)$$

Numerically: include $T_{m,0}(t)$ in a stiff `solve_ivp(method="Radau")` or BDF state. A regular power of $T_{m,0}$ may be used internally at a cold start.

Step 2: gray storage and the base front — one running time quadrature

Calculate

$$R(t) = \frac{a_{\text{rad}} T_s^4(t)}{f \rho^{1-\mu} T_{m,0}^\beta(t)}. \quad (14.7)$$

The gray storage factor is equivalently

$$G(R) = (4 + \alpha - \beta) \int_0^1 \frac{\xi^{3+\alpha-\beta}}{1 + R \xi^{4-\beta}} d\xi, \quad (14.8)$$

$$G(R) = {}_2F_1\left(1, \frac{4 + \alpha - \beta}{4 - \beta}; 1 + \frac{4 + \alpha - \beta}{4 - \beta}; -R\right). \quad (14.9)$$

The defining Euler integral is

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a} du. \quad (14.10)$$

Numerically: use `scipy.special.hyp2f1`; validate once against `quad`.

Advance the scalar history integral appearing in

$$x_{F,0}^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C T_{m,0}^{-\beta}(t) \int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'. \quad (14.11)$$

Numerically: the integral is one running quadrature state advanced by the same ODE solver.

Step 3: boundary curvature and, for a bath drive, the surface-temperature ODE

For prescribed $T_s(t)$,

$$B(t) = \frac{\varepsilon}{2} \left[1 - \frac{2 + \varepsilon}{1 - \varepsilon} \frac{\int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'}{T_s^4(t) T_{m,0}^\alpha(t) G(R(t))} \times \frac{d}{dt} \ln T_s^{4+\alpha}(t) \right]. \quad (14.12)$$

Numerically: evaluate the derivative of the prescribed `PchipInterpolator`.

For prescribed $T_{\text{bath}}(t)$, the current $T_s(t)$ is an ODE state. Using the current $T_{m,0}(t)$, current base-front history, and current $x_{F,0}(t)$, evaluate

$$B(t) = 1 - \frac{3(4 + \alpha - \beta) x_{F,0}(t) \rho^{1+\lambda}}{8g T_{m,0}^\alpha(t)} \left[\frac{T_{\text{bath}}^4(t)}{T_s^4(t)} - 1 \right], \quad (14.13)$$

then return the additional ODE right-hand side

$$\dot{T}_s(t) = \frac{2T_s^5(t)T_{m,0}^\alpha(t)G(R(t))}{\varepsilon(4+\alpha)\frac{2+\varepsilon}{1-\varepsilon}\int_0^t T_s^4(t')T_{m,0}^\alpha(t')G(R(t'))dt'} \times \left\{ \frac{3(4+\alpha-\beta)x_{F,0}(t)\rho^{1+\lambda}}{8gT_{m,0}^\alpha(t)} \left[\frac{T_{\text{bath}}^4(t)}{T_s^4(t)} - 1 \right] - 1 + \frac{\varepsilon}{2} \right\}. \quad (14.14)$$

Numerically: integrate this equation in the same stiff `solve_ivp` call. Only $T_{\text{bath}}(t)$, not its derivative, is needed.

Step 4: profile moments — local one-dimensional quadratures

For any p and z , define

$$M_p(t, z) = \int_0^1 [(1-y)(1+B(t)y)]^{p/(4+\alpha-\beta)} e^{-zy} dy, \quad (14.15)$$

$$N_p(t, z) = \int_0^1 y[(1-y)(1+B(t)y)]^{p/(4+\alpha-\beta)} e^{-zy} dy. \quad (14.16)$$

Their exact Humbert forms are

$$M_p(t, z) = \frac{1}{1 + \frac{p}{4+\alpha-\beta}} \Phi_1 \left(1, -\frac{p}{4+\alpha-\beta}; 2 + \frac{p}{4+\alpha-\beta}; -B(t), -z \right), \quad (14.17)$$

$$N_p(t, z) = \frac{1}{\left(1 + \frac{p}{4+\alpha-\beta}\right) \left(2 + \frac{p}{4+\alpha-\beta}\right)} \Phi_1 \left(2, -\frac{p}{4+\alpha-\beta}; 3 + \frac{p}{4+\alpha-\beta}; -B(t), -z \right), \quad (14.18)$$

where

$$\Phi_1(A, B; C; x, z) = \frac{\Gamma(C)}{\Gamma(A)\Gamma(C-A)} \int_0^1 e^{zu} u^{A-1} (1-u)^{C-A-1} (1-xu)^{-B} du. \quad (14.19)$$

Numerically: SciPy has no direct Humbert- Φ_1 routine. Evaluate M_p and N_p with `quad` from `scipy.integrate`; these are local profile integrals, not running time integrals and not a spatial mesh.

Step 5: areal material energy and material lag — one scalar ODE plus one local root

The evolving quantity is $U_m(t)$, not $s(t)$. At every ODE stage, the solver supplies the current trial value of $U_m(t)$. Determine the current $s(t)$ from

$$M_\beta(t, \beta s(t)) = \frac{U_m(t)}{x_{F,0}(t) f \rho^{1-\mu} T_{m,0}^\beta(t)}. \quad (14.20)$$

The root is unique and is found with `brentq`. Substitute that current root immediately into

$$\dot{U}_m(t) = \frac{a_{\text{rad}} c}{g'} \rho^{1+\lambda'} x_{F,0}(t) \left[T_s^4(t) T_{m,0}^{-\alpha'}(t) M_{4-\alpha'}(t, -\alpha' s(t)) - T_{m,0}^{4-\alpha'}(t) M_{4-\alpha'}(t, (4-\alpha')s(t)) \right]. \quad (14.21)$$

This is a closed ODE evaluation: current $U_m \rightarrow$ current scalar root $s \rightarrow$ current $\dot{U}_m$. No unknown future value of U_m is used.

A convenient first-positive-drive initial condition is $s = 0$, namely

$$U_m(t_{\text{start}}) = x_{F,0}(t_{\text{start}}) f \rho^{1-\mu} T_{m,0}^\beta(t_{\text{start}}) M_\beta(t_{\text{start}}, 0). \quad (14.22)$$

Step 6: final front — two running time quadratures

For any scalar s , define

$$\mathcal{E}_1(t, s) = a_{\text{rad}} T_s^4(t) N_4(t, 0) + f \rho^{1-\mu} T_{m,0}^\beta(t) N_\beta(t, \beta s), \quad (14.23)$$

$$\mathcal{D}(t, s) = T_s^4(t) T_{m,0}^\alpha(t) \{ [1 - B(t)] M_\beta(t, \alpha s) + 2B(t) N_\beta(t, \alpha s) \}. \quad (14.24)$$

Advance the numerator and denominator histories in

$$x_F^2(t) = x_{F,0}^2(t) \frac{\mathcal{E}_1(t, 0)}{\mathcal{E}_1(t, s(t))} \frac{\int_0^t \mathcal{D}(t', s(t')) dt'}{\int_0^t \mathcal{D}(t', 0) dt'}. \quad (14.25)$$

Numerically: the two time integrals are two additional scalar quadrature states.

Step 7: final profiles — algebraic output

For $0 \leq x \leq x_F(t)$,

$$T_r(x, t) = T_s(t) \left[\left(1 - \frac{x}{x_F(t)} \right) \left(1 + B(t) \frac{x}{x_F(t)} \right) \right]^{1/(4+\alpha-\beta)}, \quad (14.26)$$

$$T(x, t) = T_{m,0}(t) \left[\left(1 - \frac{x}{x_F(t)} \right) \left(1 + B(t) \frac{x}{x_F(t)} \right) \right]^{1/(4+\alpha-\beta)} \times \exp \left[-s(t) \frac{x}{x_F(t)} \right]. \quad (14.27)$$

Set both temperatures to zero beyond the sharp front.

LTE checks

On the LTE manifold,

$$T_{m,0}(t) = T_s(t), \quad s(t) = 0, \quad (14.28)$$

the final correction is unity and $x_F = x_{F,0}$. Retaining $G(R)$ gives an LTE HR model with radiation-energy storage. If additionally $R \ll 1$, then $G(R) \rightarrow 1$ and the classical LTE Hammer–Rosen front is recovered:

$$x_F^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C T_s^{-\beta}(t) \int_0^t T_s^{4+\alpha}(t') dt'. \quad (14.29)$$

In a Marshak problem, $T_{\text{bath}} = T_s$ implies zero net flux, not LTE and not a prescribed-surface boundary.

For an LTE bath drive, Cohen–Heizler Appendix A determines the unknown $T_s(t)$ from the global HR areal-energy relation together with $\dot{E}_{\text{HR}} = (a_{\text{rad}}c/2)(T_{\text{bath}}^4 - T_s^4)$. The direct bath equation in Step 3 instead uses the local HR surface gradient. It has the same physical Marshak boundary and a consistent LTE limit, but it is a different simplified closure and is not claimed to be algebraically identical to Appendix A.

Complete numerical workflow and cost

Prescribed $T_s(t)$. At each `solve_ivp` stage:

1. read $T_s(t)$ and its interpolant derivative;
2. read the current states $T_{m,0}(t)$ and $U_m(t)$ and the three running time integrals;
3. calculate R , $G(R)$, $x_{F,0}$, and B ;
4. solve the current s root from the current U_m ;
5. evaluate the local M_p, N_p profile integrals;
6. return $\dot{T}_{m,0}$, $\dot{U}_m$, and the three time-integral derivatives.

The numerical content is

$$\boxed{2 \text{ scalar physical ODEs} + 3 \text{ running time quadratures} + 1 \text{ local scalar root.}} \quad (14.30)$$

Prescribed $T_{\text{bath}}(t)$. At each stage, the current $T_s(t)$ is supplied by the ODE state; it is not precomputed. Read $T_{\text{bath}}(t)$, evaluate the common quantities with the current $T_s(t)$, return $\dot{T}_s(t)$ from Step 3, and then perform the same s root, material-energy, and transport operations. The numerical content is

$$\boxed{3 \text{ scalar physical ODEs} + 3 \text{ running time quadratures} + 1 \text{ local scalar root.}} \quad (14.31)$$

Both branches additionally evaluate a fixed small number of one-dimensional profile integrals with `quad`. Their cost and state dimension do not grow with spatial resolution. There is no spatial grid, diffusion matrix, nonlinear PDE iteration, or time-dependent integral equation.